

2D pore-scale simulation of wide-band electromagnetic dispersion of saturated rocks

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ABSTRACT

Effective medium theories (EMTs) are invoked routinely to interpret multifrequency dispersions of dielectric permittivity and electrical conductivity of saturated rocks. However, EMTs exhibit limitations that substantially restrict their validity for petrophysical interpretation. For instance, pore connectivity is of significant interest in the study of subsurface reservoirs, but no existing EMT includes it as an explicit property in the analysis of kilohertz- to gigahertz-range dielectric measurements. We introduce a new approach to quantify the effects of pore geometry and connectivity on the kilohertz-gigahertz frequency dispersion of dielectric permittivity and electrical conductivity of clay-free porous rocks. This approach is based on the numerical solution of the internal electric fields within submicron-resolution pore maps constructed with grain and rock pixels. The discrepancy between the internal fields and electrical currents calculated for a

homogeneous scatterer and those calculated for a given pore map is minimized to yield the effective electrical conductivity and dielectric constant for that pore map. This minimization is performed independently for each frequency and is verified to agree implicitly with Kramers-Kronig's causality relationships. We show that EMTs only predict an average dispersion for given microscopic geometrical parameters (e.g., porosity, pore eccentricity), whereas individual realizations honoring the same parameters are associated with dispersion about average values predicted by EMTs. Unlike any EMT prediction, we show that pore connectivity plays a major role in both the shape and amplitude of wide-band electromagnetic property dispersions. The simulation procedure introduced in this paper provides a systematic method to assess the sensitivity of a multitude of pore-scale properties on the macroscopic wide-band dielectric dispersion of saturated rocks.

INTRODUCTION

Maxwell-Wagner interfacial polarization process

The complex electrical impedance measured across a saturated rock is in some ways analogous to a parallel resistor/capacitor model. Geometrically normalized measurements of conductance and capacitance yield the effective conductivity σ_{eff} and dielectric permittivity ϵ_{eff} of the rock sample, both real quantities. Electrical conductivity quantifies dissipation of energy, whereas dielectric permittivity quantifies energy storage. If the rock sample is regarded as a conductor of complex conductivity $\bar{\sigma}^*$ subject to monochromatic excitation of frequency f and radian frequency $\omega = 2\pi f$, $\bar{\sigma}^*$ can be expressed in terms of the measured values of σ_{eff} and ϵ_{eff} with the expression

$$\bar{\sigma}^* = \sigma_{\text{eff}} - i\omega\epsilon_{\text{eff}}, \quad (1)$$

where the $e^{-i\omega t}$ time harmonic convention is adopted for the exciting electric field and t is time. Similarly, if the rock sample is regarded as a capacitor of complex dielectric permittivity $\bar{\epsilon}^*$ under the same conditions, $\bar{\epsilon}^*$ is equal to

$$\bar{\epsilon}^* = \frac{i\bar{\sigma}^*}{\omega} = \epsilon_{\text{eff}} + \frac{i\sigma_{\text{eff}}}{\omega}. \quad (2)$$

The dielectric constant κ of the medium is the ratio of effective dielectric permittivity to that of vacuum, $\epsilon_0 = 8.854 \times 10^{-12}$ F/m. Conductivities and permittivities of pure charge-free isolated rocks and bulk hydrocarbons are constant over the entire kilohertz-gigahertz frequency range. For increasing values of frequency, salty wa-

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ter begins to exhibit frequency dependence near 1 GHz (for simplicity, we assume that the dielectric properties of water remain constant over the entire kilohertz-gigahertz frequency range).

In the case of rock-fluid mixtures, electric charges accumulate at the interface between brine and rock (or oil). Within these charged surfaces, brine polarizes in the form of a macroscopic dipole, which gives rise to frequency-dependent macroscopic polarization. At low frequencies, macroscopic dipoles reach equilibrium before the incident field has changed notably, giving rise to polarization buildup. When frequency increases, the orientation of macroscopic dipoles cannot follow the applied field because of the viscosity of the fluid, thereby resulting in energy dissipation, increased electrical conductivity, and reduced dielectric permittivity.

This behavior, summarized by Bona et al. (1998, 2002), is known as the Maxwell-Wagner phenomenon; it describes the Debye relaxation behavior of complex effective permittivity of composite material as a function of frequency, namely,

$$\bar{\epsilon}^*(\omega) = \bar{\epsilon}^*(\omega \rightarrow \infty) + \frac{\Delta\bar{\epsilon}^*}{1 + i\omega\tau}. \quad (3)$$

In this equation, $\Delta\bar{\epsilon}^*$ is the difference between the asymptotic values of the composite permittivity at zero and at infinite frequencies, and τ is the dielectric relaxation time of the equivalent dipole formed by brine pores. The relaxation time τ is a complex function of pore-shape factor, pore isotropic orientation, and rock/brine permittivities (Sillars, 1937). Variants of equation 3 exist, with adjustable exponents introduced to match experimental data (Bona et al., 2002). Other models adapt equation 3 for distributions of pore geometries (Lysne, 1983) that include several inflection points in a single dielectric dispersion curve.

Dielectric mixing laws and other polarization models

Dielectric mixing laws based on effective medium theory (EMT) and polarization averaging also yield sigmoidal frequency-dispersion curves. Work by Choy (1999) and Sihvola (2002) summarizes the assumptions and limitations of existing dielectric EMTs.

Starting from Maxwell-Garnett's mixing formula, which is valid strictly for isolated inclusions, a family of models was developed to describe the behavior of dense composites. These mixing laws include Bruggeman's model, coherent potential, Looyenga's model, Sen-Scala-Cohen's model (Sen et al., 1981), and Mendelson-Cohen's (1982) effective medium approximation (EMA). In particular, Choy (1999) provides a thorough description of the Maxwell-Garnett and Bruggeman averages from a theoretical viewpoint and compares the two approaches and their corresponding limitations. In similar fashion to Maxwell-Wagner descriptions, effective medium models account for (1) contrasts of electrical properties between the components and (2) eccentricities and anisotropic distributions of brine-filled zones (or, similarly, of grains and resistive oil blobs), regardless of pore or grain size.

Sihvola (2002) also describes a family of power-law dielectric models where the effective complex permittivity of the mixture taken to some power, $(\bar{\epsilon}^*)^{1/N}$, is equal to the volumetric average of $(\bar{\epsilon}^*)^{1/N}$ for each component. These models include the popular complex refractive index method (CRIM) with $N = 2$, which has been widely applied as a convenient phase volumetric average in the gigahertz range. CRIM also forms the basis for ongoing efforts to include

grain morphology in predicting the dielectric constant of inhomogeneous carbonate rocks in the 100-MHz to 1-GHz frequency range (Seleznov et al., 2006).

These mixing laws shed much theoretical insight on the relaxation processes taking place in saturated rocks. Unfortunately, they provide little practical petrophysical information either because they fail to fit laboratory measurements accurately (as in Kenyon, 1984) or because they use parameters that are difficult to determine (e.g., those of the Sillars [1937] model) without direct observation (or spatial imaging) of the sample. Their assumptions and physical consistency are also sometimes questionable (Sihvola, 2002), leaving one to wonder which mixing laws are applicable to which specific cases, if any. Díaz et al. (1998) provide a 14-point summary of both conceptual and practical issues related to applying dielectric mixing laws. A major failure of these mixing laws based on electric-field averaging is that they implicitly neglect both the duality of Maxwell's equations and the electromagnetic (EM) coupling between brine bodies within rocks. This limitation is overcome by del Río and Whitaker (2000a, 2000b), who enforce the dual averaging of electric and magnetic fields simultaneously and mathematically demonstrate the limits of single-field averaging results. A similar duality is at the basis of the numerical approach we develop for calculating the effective conductivity and dielectric permittivity of digital composite media, as discussed in a subsequent section.

Finally, attention is called to the remarkable but rarely mentioned Stroud-Milton-De (SMD) analytical model (Stroud et al., 1986), based on Bergman's analytic model for a two-component composite. The strength of the SMD model is that it captures the pore-space geometrical information as a cascade of gamma functions of porosity and DC conductivity as long as the contact area between grains remains smaller than that of rock/fluid interfaces. The generality of the SMD model is supported by excellent data fitting with dielectric dispersions measured on a variety of rock samples between 10 MHz and 1 GHz. A similar analytical approach based on the single-mode decomposition of the rock dielectric permittivity spectrum has been developed by Díaz et al. (1998), who capture the geometry of a dielectric composite with a single percolation threshold. The main problem that remains is to assign the right percolation threshold to the rock sample for proper geometrical description.

Objectives and methodology

The analytical models described above suggest strong ties between grain and pore geometry, DC conductivity, and percolation ability of the water phase within the rock. However, much work remains to relate dielectric frequency dispersions with pore connectivity at the small scale and with rock hydraulic permeability at the large scale. Therefore, it is appropriate to quantify the respective contributions of pore or grain eccentricity on dielectric dispersion as collective objects or clusters versus the contribution of pore connectivity regardless of eccentricity at a macroscopic scale.

The objective of this paper is twofold: (1) to develop a numerical method that enables the accurate simulation of dielectric dispersion effects at the pore scale on a frequency range wider than allowed by existing methods and (2) to quantify the influence of explicit pore connectivity on the dielectric dispersion of 2D digital rock models. We specifically target our analysis to the 10 kHz–1 GHz range, which corresponds to the frequency band of borehole measure-

ments acquired with open-hole logging resistivity and induction (10–200 kHz), logging-while-drilling propagation (2 MHz), and EM propagation (1 GHz) instruments.

We introduce a numerical framework consisting of 2D digital pore maps that distribute submicrometer brine and rock/oil pixels of variable complex conductivities included in realistic pore and throat geometries. Frequency-dependent dispersions of upscaled effective conductivity σ_{eff} and dielectric constant κ_{eff} for disc-shaped pore maps are estimated based on EM propagation principles. Specifically, the electric fields induced by a monochromatic EM excitation and the corresponding internal currents are computed within the pore maps and compared to those that would be excited by the same incident plane wave in a homogeneous cylindrical scatterer of given conductivity and dielectric constant. The conductivity and dielectric constant of the homogeneous cylinder that minimize the misfit between the internal fields and electrical currents in the pore map and in the homogeneous medium become the best-fit effective properties of the pore map. Without loss of generality, our developments assume macroscopic isotropy of the rock models when using an isotropic homogeneous cylinder to calculate the effective values of σ_{eff} and κ_{eff} .

NUMERICAL METHOD FOR CALCULATING EFFECTIVE CONDUCTIVITY AND DIELECTRIC PERMITTIVITY

Scatterer geometry

The geometrical configuration adopted for the pixel-based scatterer is compatible with an existing fast analytical solution of EM scattering within a homogeneous dielectric and lossy cylinder on a wide frequency range (kilohertz to gigahertz). Pekonen et al. (1999) apply similar principles to simulate the megahertz-range response of 2D pixel-based purely dielectric mixtures bounded by a waveguide. These authors compute the effective dielectric constant of the medium from the reflection and transmission coefficients at the scatterer boundaries. The main disadvantage of the waveguide method is that it remains accurate only below 80 MHz (Pekonen et al., 1999), whereas our problem demands high numerical accuracy for frequencies as high as 1 GHz.

To circumvent this limitation, we introduce a new estimation approach based on the cylindrical scatterer geometry shown in Figure 1. Without restriction on frequency, the medium is described as a disc populated with square pixels of contrasting conductivities and dielectric constants of grain and fluids representing the pore map. We consider transverse magnetic (TM) excitation such that the vector components of the incident electric field are in the same plane defined by the cross section of the 2D scatterer. This TM polarization causes the induced electrical currents to cross boundaries between pixels with contrasts of electric and dielectric properties.

The presence of corners, vertices, and irregular shapes on the boundary of the disc-shaped scatterer can cause significant fringe effects on the internal electric field. We choose a circular boundary to define the spatial extension of the pore map to mitigate this problem, to take advantage of the closed-form analytical resolution of the electric fields in a cylindrical scatterer (Balanis, 1989), and ultimately to compare simulations against measurements performed on cylindrical core data. In addition, a high-resolution pore map is necessary to attenuate boundary effects at low frequencies for the case of resistive pixels. As shown in Figure 2, at the nanometer- to micrometer-scale resolution used in our models, pore maps with resolution

less than 400×400 pixels entail substantial boundary fringing below 10 MHz for a homogeneous distribution of resistive pixels and 1 S/m background water. These numerical artifacts bias the estimation of effective properties of the scattering medium. In what follows, we show that spatial discretizations of the scattering medium

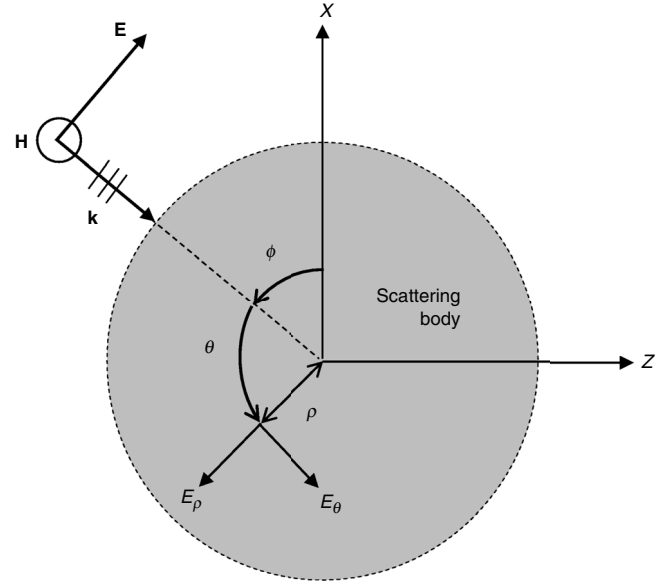


Figure 1. Model geometry. The incident transverse magnetic (TM) polarized plane wave illuminates the 2D scattering medium at an angle ϕ with respect to the x -axis. $\mathbf{E}$: incident electric field. $\mathbf{H}$: incident magnetic field. $\mathbf{k}$: wave (propagation) vector. The polar coordinates of a point with respect to the incident plane wave and the scatterer centers are (ρ, θ) , while the polar projections of the electrical field at that point are E_ρ, E_θ .

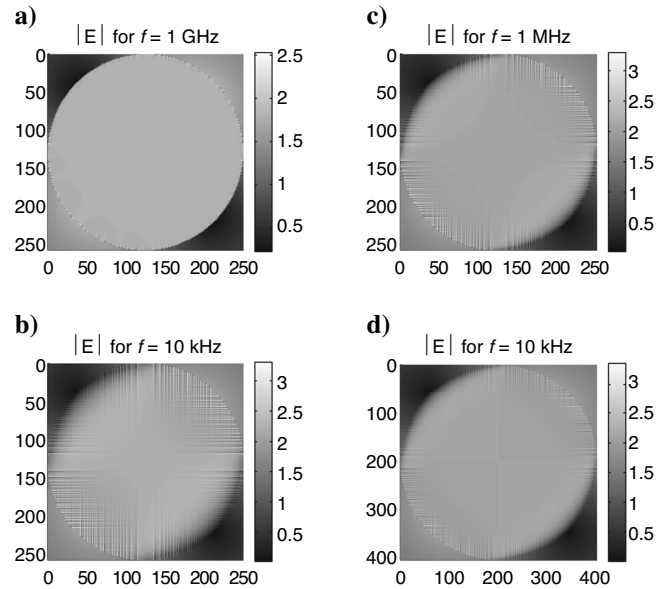


Figure 2. Illustration of the impact of frequency and map resolution on the boundary fringing that distorts the internal electric fields computed within a resistive homogeneous disc. All maps include 255×255 -pixel resolution except for panel (d) (401×401 -pixel resolution). Electric excitation is in the form of an incident TM wave illuminating the disc in a top-right to bottom-left direction.

larger than 400×400 pixels enable reliable estimations of effective properties of a homogeneous disc in the entire kilohertz-gigahertz range.

Internal electric fields

We assume that the individual pixel conductivities as well as the conductivity of the lossy brine background are isotropic. Equation 1 defines the complex effective conductivity for each pixel. Similarly, the complex effective conductivity of the water background is given by

$$\bar{\sigma}_b^* = \sigma_b - i\omega\epsilon_0\kappa_b, \quad (4)$$

where σ_b is the brine effective conductivity and κ_b is its dielectric constant. If $\mu_0 = 1.257 \times 10^{-6}$ H/m is the vacuum magnetic permeability, the wavenumber k_b of the background is defined as

$$k_b^2 = i\omega\mu_0\bar{\sigma}_b^*. \quad (5)$$

The pore map is regarded as a complex scatterer where each element interacts with the incident wave and where the cross-coupled response (or cross scattering) of all the elements gives rise to the overall EM response of the saturated rock. The fast Fourier transform method of moments (FFT-MoM) (Wang, 1991; Catedra et al., 1995; Avdeev et al., 2002) was implemented in two dimensions to solve Maxwell's equations and compute the complex electric-field values in each pixel of the pore map. We selected this method over other numerical methods for three reasons: (1) it is formulated in the same frequency domain as the closed-form analytical homogeneous solution used for effective property inversion, (2) it explicitly honors the scatterer boundaries and does not require implementing special boundary conditions, and (3) it accurately and efficiently incorporates large contrasts in material electrical properties (as encountered at the water/grain interface).

All calculations are performed in the (x, z) 2D Cartesian coordinate frame described in Figure 1. We solve the general integral equation numerically for the electric field $\mathbf{E}$ at location $\mathbf{r} = (x, z)$ given by

$$\begin{aligned} \mathbf{E}(\mathbf{r}) = \mathbf{E}_b(\mathbf{r}) + \int_{\mathbf{r}' \in V_{\text{scat}}} i\omega\mu_0\vec{\mathbf{G}}(\mathbf{r}, \mathbf{r}') \\ \cdot \Delta\boldsymbol{\sigma}^*(\mathbf{r}') \cdot \mathbf{E}(\mathbf{r}') dV, \end{aligned} \quad (6)$$

where $\mathbf{E}_b$ is the background field normalized by the $e^{-i\omega t}$ time-harmonic variation imposed by a TM incident plane wave that forms a counterclockwise angle ϕ with the x -axis, i.e.,

$$\mathbf{E}_b(\mathbf{r}) = \begin{bmatrix} -e^{ik(-x \cos \phi + z \sin \phi)} \sin \phi \\ 0 \\ -e^{ik(-x \cos \phi + z \sin \phi)} \cos \phi \end{bmatrix}. \quad (7)$$

The scatterer consists of rock and fluid pixels of effective complex conductivity $\bar{\sigma}^*$ occupying the volume V_{scat} . We solve equation 6 as the Riemann sum of the scattering caused by all the pixels forming V_{scat} . For each pixel, $\Delta\boldsymbol{\sigma}^*$ is the electrical conductivity contrast tensor between the scatterer and the water background. The tensor $\Delta\boldsymbol{\sigma}^*$ is assumed to be isotropic at the pixel level and, following equations 1 and 4, becomes the complex scalar $\Delta\bar{\sigma}^* = \bar{\sigma}^* - \bar{\sigma}_b^*$. Finally, in equation 6, $\vec{\mathbf{G}}(\mathbf{r}, \mathbf{r}')$ is the dyadic Green's tensor for the electric field. This tensor quantifies the cross interaction of two scatterers located

in the (x, z) plane of the pore map at positions $\mathbf{r}$ and $\mathbf{r}'$, respectively, and is defined as the solution of the dyadic wave equation,

$$\nabla \times \nabla \times \vec{\mathbf{G}}(\mathbf{r}, \mathbf{r}') - k_b^2 \vec{\mathbf{G}}(\mathbf{r}, \mathbf{r}') = \vec{\mathbf{I}}\delta(\mathbf{r} - \mathbf{r}'), \quad (8)$$

i.e.,

$$\vec{\mathbf{G}}(\mathbf{r}, \mathbf{r}') = \left(\vec{\mathbf{I}} + \frac{1}{k_b^2} \nabla \nabla \right) g(\mathbf{r}, \mathbf{r}'). \quad (9)$$

In equation 9, $\vec{\mathbf{I}}$ is the identity tensor, $\nabla \mathbf{r}$ is Dirac's delta function, and g is the scalar Green's function that satisfies the wave equation with a point source excitation:

$$\nabla^2 g(\mathbf{r}, \mathbf{r}') + k_b^2 g(\mathbf{r}, \mathbf{r}') = -\delta(\mathbf{r} - \mathbf{r}'). \quad (10)$$

In the process of discretizing equation 6, $\vec{\mathbf{G}}(\mathbf{r}, \mathbf{r}')$ is integrated on the rectangular area of each pixel of the pore map using the analytical expressions derived by Gao et al. (2005). Appendix A quantifies the accuracy with which the fields are determined within a homogeneous cylinder when using this method of solution.

Figure 3 shows an example of internal currents distributed within a pore map that consists of brine ellipses ($\sigma_w = 1$ S/m, $\kappa_w = 70$) embedded in a rocklike host ($\sigma_{\text{host}} = 0$, $\kappa_{\text{host}} = 2$) for two cases of perpendicular plane-wave TM excitation. At 100 MHz, the out-of-phase dielectric effects are more significant than the in-phase conductivity effects. Consequently, the internal currents induced across the scatterer are normal to the incoming electric field, i.e., parallel to the incoming TM wave according to the nomenclature of Figure 1. A variation of the scale of the current amplitudes in Figure 3 emphasizes two main phenomena. First, the distribution of low-current amplitudes (middle panels) shows the directionality of the internal currents and describes how the currents bypass the ellipsoidal inclusions that block the passage of the incident waves. Second, the distribution of high-current amplitudes (right-hand panels) shows that electrical current reaches a maximum in the brine inclusions that are normal to the incident waves (parallel to the incident electric fields), while the amplitude of currents decreases in the inclusions parallel to the incident wave.

Inversion of effective conductivity and dielectric permittivity

From the viewpoint of EM propagation, we define the effective conductivity and dielectric constant of the composite pore map as the conductivity σ_h and dielectric constant κ_h of a homogeneous and isotropic scatterer that reproduces as closely as possible the scattered field attributable to the pore map. These conditions include angle of incidence, frequency and amplitude of the exciting EM wave, and scatterer shape. Hence, the estimation of σ_h and κ_h is consistent with Maxwell's equations and can be approached as a minimization problem.

It follows from equation 6 that the electrical field propagated to any location in space is a weighted function of the currents $\mathbf{J}$ inside the scatterer, i.e.,

$$\mathbf{J} = \Delta\bar{\sigma}^* \mathbf{E}. \quad (11)$$

Therefore, if one seeks to minimize the discrepancy between the electric fields scattered by the pore map and the fields scattered by a homogeneous scatterer regardless of the position where the fields are evaluated, the same results should be obtained if one minimizes the discrepancy of internal currents $\mathbf{J}$ within the scatterers.

If the pore map is considered as the scatterer, $\Delta\bar{\sigma}^* = \bar{\sigma}^* - \bar{\sigma}_b^*$ varies for each pixel. For a homogeneous scatterer of real conductivity σ_h and dielectric constant κ_h , we define its complex conductivity as $\bar{\sigma}_h^* = \sigma_h - i\omega\epsilon_0\kappa_h$; the conductivity contrast with respect to the background becomes $\Delta\bar{\sigma}^* = \bar{\sigma}_h^* - \bar{\sigma}_b^*$ for the entire scatterer. We minimize the discrepancy of internal currents $\mathbf{J}$ over the entire scatterer by minimizing the cost function:

$$\begin{aligned}\Psi &= \|(\bar{\sigma}^* - \bar{\sigma}_b^*)\mathbf{E} - (\bar{\sigma}_h^* - \bar{\sigma}_b^*)\mathbf{E}_h\|_{L_2} \\ &= \|\bar{\sigma}^*\mathbf{E} - \bar{\sigma}_h^*\mathbf{E}_h - \bar{\sigma}_b^*(\mathbf{E} - \mathbf{E}_h)\|_{L_2},\end{aligned}\quad (12)$$

where $\mathbf{E}$ is the total electric field at a given pixel of complex conductivity $\bar{\sigma}^*$ and $\mathbf{E}_h$ is the electric field analytically calculated for the homogeneous scatterer of complex conductivity $\bar{\sigma}_h^* = \sigma_h - i\omega\epsilon_0\kappa_h$. Because the pore map does not exceed 1 mm², this analytical value is identical across the entire disc, and we calculate it close to the disc center. The L_2 norm adopted for the cost function Ψ enforces the quadratic misfit over the entire surface of the pore map. Therefore, the values of σ_h and κ_h for which Ψ reaches its minimum correspond to the effective, upscaled electrical properties σ_{eff} and κ_{eff} of the pore map for this frequency and this angle of excitation.

Appendix B describes the practical implementation of the minimization of equation 12 with a disc-shaped, homogeneous, isotropic 2D scatterer. In the same appendix, we emphasize that the choice of the cost function,

$$\Psi_3 = \|\bar{\sigma}^*\mathbf{E} - \bar{\sigma}_h^*\mathbf{E}_h\|_{L_2} + |\bar{\sigma}_b^*| \times \|\mathbf{E} - \mathbf{E}_h\|_{L_2}, \quad (13)$$

provides the widest range of accuracy, including rock conductivities as low as 1 mS/m.

Our approach for estimating σ_{eff} and κ_{eff} is performed frequency by frequency independently. As shown in the next sections, it is

remarkable that our estimation method produces smooth dielectric dispersion curves that implicitly satisfy the Kramers-Kronig (KK) relationships (Landau and Lifschitz, 1960).

CONNECTIVITY EFFECTS IN DILUTE CONCENTRATIONS OF BRINE INCLUSIONS

Effective medium approximation with depolarization factor l

Mendelson and Cohen's (1982) EMA (subsequently adapted by Feng and Sen [1985] to the case of multiphase fluid saturations) is one of the few mixing laws applicable to 2D media and that explicitly integrate grain or pore geometry information over a wide frequency range. An often-overlooked property of the EMA in its application to interpret measurements of saturated rocks is that this approximation remains strictly valid for dilute concentrations of impurities (in our case, lossy dielectric brine inclusions embedded in a resistive rock host).

The EMA assumes an incremental refinement of the rock/water mixture where infinitesimal amounts of rock are added to the rock/water mixture of a previous step and where the new complex conductivity of the mixture is estimated by averaging the electric fields and the electric displacements throughout the entire volume. At each step, the rock grains added to the mixture are oblong spheroids randomly oriented and individually characterized by the same geometric depolarization factor l . The EMA then yields a relationship between the porosity ϕ of the saturated rock and the complex dielectric constants of water $\bar{\epsilon}_w^*$, rock grains $\bar{\epsilon}_{\text{grain}}^*$, and mixture $\bar{\epsilon}^*$ (Mendelson and Cohen, 1982; Kenyon, 1984; Sen, 1984), given by

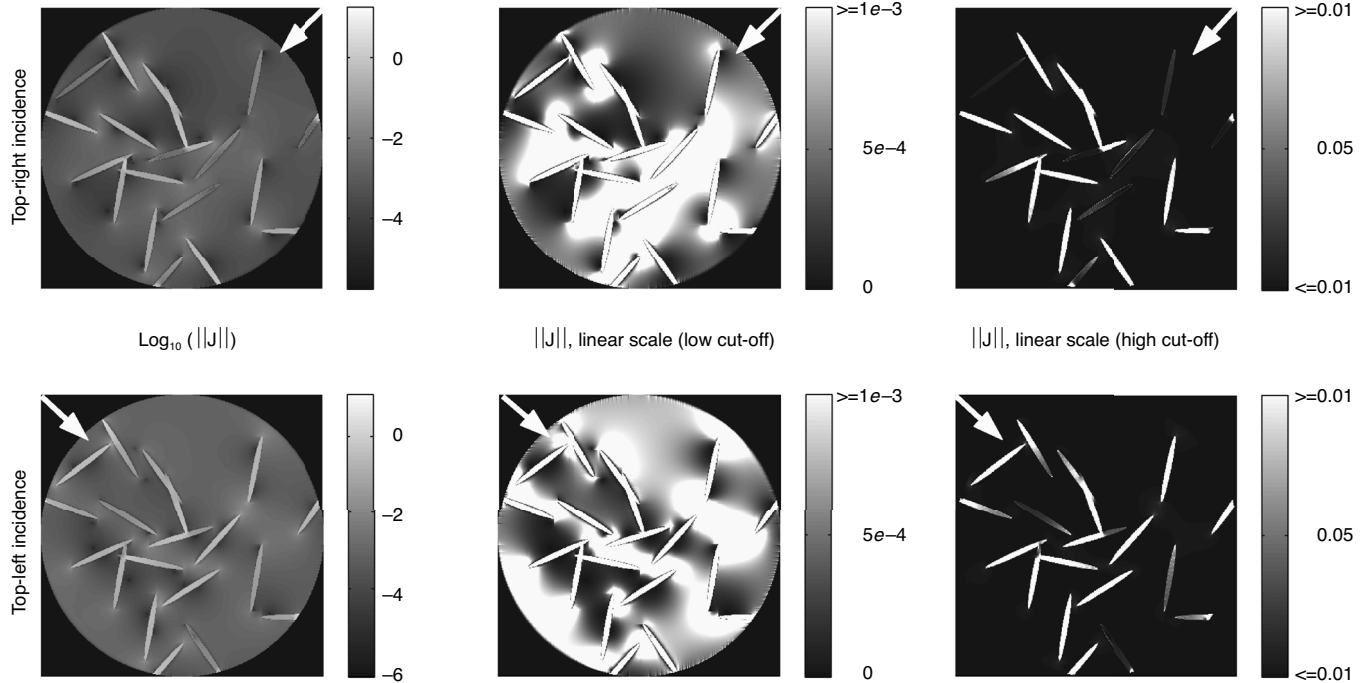


Figure 3. Graphical description of internal current amplitudes within pore map B in Figure 4 at 100 MHz for two perpendicular angles of incidence and at different scales. The arrows indicate the directions of plane-wave incidence.

$$\ln \phi = \frac{3l(1-l)}{1+3l} \ln \left(\frac{\bar{\epsilon}_w^*}{\bar{\epsilon}^*} \right) + \ln \left(\frac{\bar{\epsilon}^* - \bar{\epsilon}_{\text{grain}}^*}{\bar{\epsilon}_w^* - \bar{\epsilon}_{\text{grain}}^*} \right) + \frac{2(1-3l)^2}{(1+3l)(5-3l)} \times \ln \left[\frac{(5-3l)\bar{\epsilon}_w^* + (1+3l)\bar{\epsilon}_{\text{grain}}^*}{(5-3l)\bar{\epsilon}^* + (1+3l)\bar{\epsilon}_{\text{grain}}^*} \right]. \quad (14)$$

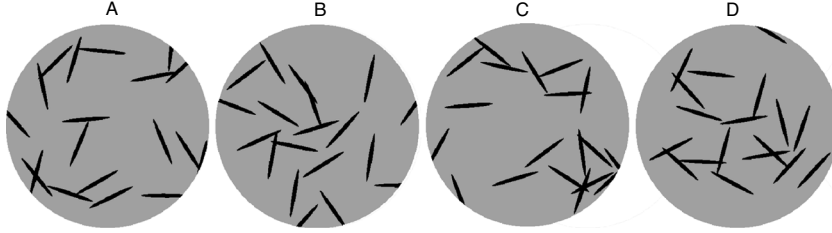


Figure 4. Pore maps exhibiting 8% porosity and measuring 401×401 pixels. The brine inclusions consist of 4×48 pixels water ellipses (black) embedded in rock host (gray).

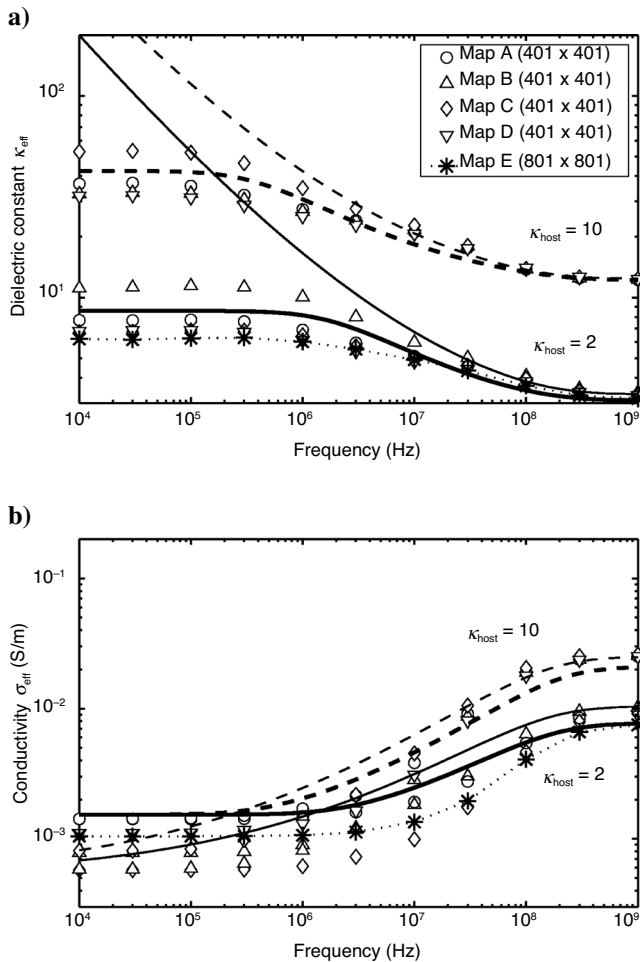


Figure 5. Frequency dispersions of (a) effective dielectric constant and (b) electrical conductivity for pore maps A–E (Figures 4 and 9). Simulation results are identified with markers for $\kappa_{\text{host}} = 2$ and $\kappa_{\text{host}} = 10$ (for map E, $\kappa_{\text{host}} = 2$ only). Corresponding mixing-law results are identified with plain curves for $\kappa_{\text{host}} = 2$ and dashed curves for $\kappa_{\text{host}} = 10$. Thick curves identify the EMA results from equation 14 that best fit all the dielectric and conductivity dispersions simultaneously (for $l = 0.82$), whereas thin curves identify the power-law results from equation 16.

The general expression for l can be found in Landau and Lifshitz (1960). For 2D elliptic inclusions of eccentricity e , l is given by the expression

$$l = \frac{\sqrt{e}}{1 + \sqrt{e}} \quad (15)$$

(Sihvola, 2005). The comparisons that follow are based on the pore maps shown in Figure 4, where ellipses of brine ($\sigma_w = 1$ S/m, $\kappa_w = 70$) are distributed randomly in an insulating rock-like host. These ellipses share unique semiaxes values equal to 4×48 pixels, thereby involving a constant eccentricity $e = 12$ and a depolarization factor $l = 0.776$. In all cases, the porosity formed by these brine inclusions is kept equal to 8% and enforces the assumption of dilute concentration of inclusions. Two possible dielectric constants are assigned to the host, $\kappa_{\text{host}} = 2$ and $\kappa_{\text{host}} = 10$, that bound realistic values for rock (4.5 for quartz and 7.5 for calcite).

Statistical significance of sigmoid dispersions

At a first stage, let us consider the random pore distributions of the maps in Figure 4. The elliptical pores are distributed randomly and exhibit different geometrical arrangements; pores in maps A and C seem more organized as local clusters, while pores in maps B and D look more homogeneously distributed across the entire space. Each map includes a distribution of 401×401 pixels with a resolution of $1 \mu\text{m}^2/\text{pixel}$. All simulations are performed for one angle of incidence and include two frequency samples per decade.

The EMA solution of equation 14 is plotted in Figure 5 with a depolarization factor of $l = 0.82$ for both values of κ_{host} . In this figure, EMA dispersions fit very well the average of both the dielectric and conductivity simulation results for maps sharing the same value of κ_{host} on the entire frequency band. Two conclusions stem from this result. First, the scattering that appears between the different pore map realizations and the EMA sigmoids suggest that sigmoid analytical mixing rules capture an average trend for the given set of porosities, material electrical properties, and l values but do not account for the geometrical features that entail variations of σ_{eff} and κ_{eff} from one pore map to another. Second, the difference between the best-match value of l (0.82) and the theoretical value of l (0.776) is small, possibly because some ellipses touch each other and form clusters with overall eccentricities larger than 12.

Consistency of simulation results

We then verify the consistency of our dilute-inclusion simulation results with respect to a volumetric dielectric power-law, brine-conductivity sensitivity, Argand plots, and KK relations. Figure 5 first confirms that simulation results obtained with $\kappa_{\text{host}} = 2$, $\kappa_w = 70$, and $\sigma_w = 1$ S/m fit a CRIM-like power-law behavior of exponent $N = 3$ above 30 MHz for both conductivity and dielectric constant, whereby

$$\bar{\epsilon}^* = [\phi(\bar{\epsilon}_w^*)^{1/3} + (1 - \phi)(\bar{\epsilon}_{\text{grain}}^*)^{1/3}]^3. \quad (16)$$

Specifically considering map A, we also verify that simulation results remain consistent when the brine conductivity is changed to

$\sigma_w = 0.1$ and 10 S/m. Figure 6 confirms the trend predicted by both EMA and the power law.

The Argand plot constructed in Figure 7 from the results of Figure 6 and from the corresponding EMA predictions also agrees very well with the expected convex curvature measured in saturated rocks (de Lima and Sharma, 1992). An Argand plot describes the evolution of reactance with respect to resistance of a sample for increasing values of frequency. Let us designate ρ' and ρ'' the real in-phase and out-of-phase resistivities, respectively. We then equate the normalized complex resistivity $\bar{\rho}^* = \rho' - i\rho''$ to the inverse of the corresponding normalized admittance $\bar{\sigma}^* = \sigma_{\text{eff}} - i\omega\epsilon_0\kappa_{\text{eff}}$ to obtain the relationships

$$\rho' = \frac{\sigma_{\text{eff}}}{(\sigma_{\text{eff}})^2 + (\omega\epsilon_0\kappa_{\text{eff}})^2} \quad \text{and} \quad \rho'' = \frac{-\omega\epsilon_0\kappa_{\text{eff}}}{(\sigma_{\text{eff}})^2 + (\omega\epsilon_0\kappa_{\text{eff}})^2}, \quad (17)$$

which are the quantities plotted in Figure 7.

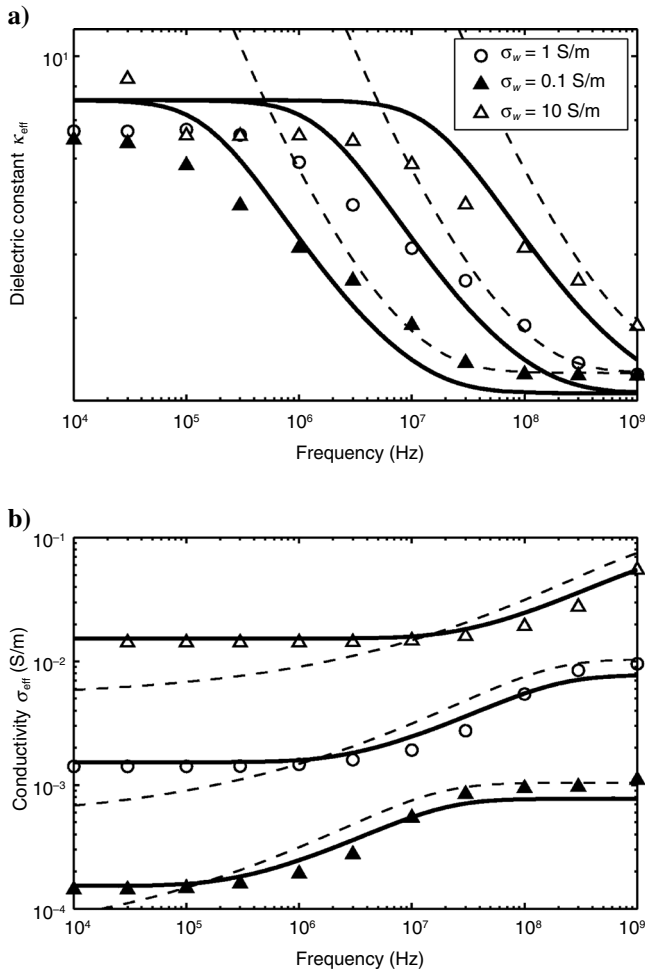


Figure 6. Frequency dispersions of (a) effective dielectric constant and (b) electrical conductivity for pore map A with $\kappa_{\text{host}} = 2$ and three magnitudes of water conductivity σ_w . Markers identify simulation results (starting with the ones plotted in Figure 5 for $\sigma_w = 1$ S/m), plain thick curves identify EMA results from equation 14 with $l = 0.82$, and dashed curves identify power-law results from equation 16.

KK relations normally are used to verify the quantitative consistency of wide-band dielectric experimental measurements. In a general sense, the current density $\mathbf{J} = \sigma^* \mathbf{E}$ and the electric displacement field $\mathbf{D} = \epsilon^* \mathbf{E}$ through a material assume that σ^* and ϵ^* are complex numbers that can be written in terms of their real and imaginary parts as $\sigma^* = \sigma' - i\sigma''$ and $\epsilon^* = \epsilon' + i\epsilon''$. As a causal physical mechanism, the KK relationship (Landau and Lifschitz, 1960) establishes the following coupling between ϵ' and ϵ'' :

$$\epsilon'(\omega) = 1 + \frac{2}{\pi} \int_0^\infty \frac{\omega_0 \epsilon''(\omega_0)}{\omega_0^2 - \omega^2} d\omega_0. \quad (18)$$

A similar relationship holds between σ' and $-\sigma''$, so that κ_{eff} at any frequency can be calculated from the dispersion of σ_{eff} in the ideal case where σ_{eff} is measured along the entire 0 to ∞ frequency range. Methods have been developed to account for the frequency-range truncation implicit in laboratory measurements (Lucarini et al., 2005). We used a companion Matlab routine (available from <http://www.mathworks.com/matlabcentral>) to calculate the dielectric constant κ_{eff} predicted from the σ_{eff} spectrum obtained from our simulations. Because the wide frequency band used in this work spans five frequency decades and because such a KK estimate uses data linearly and not logarithmically spaced in the frequency domain, for each numerical experiment, we calculated two KK estimates of dielectric constant spectra from the simulated spectra of conductivity.

A high-frequency KK dielectric spectrum was first computed for frequencies between 10 kHz and 1 GHz and interpolated every 25 kHz. As shown below, the lack of data below 1 kHz and the 25 -kHz sampling interval causes this KK-derived dielectric spectrum to be invalid below the 1 -MHz range. We therefore defined a second, low-frequency KK estimate of dielectric constant with a lower sampling increment (2.5 kHz) to interpolate the simulated conductivities below 100 MHz, assuming that the dielectric response of the composite remains constant below 10 kHz down to 10 Hz. This proxy calculation allowed us to reconstruct a consistent dielectric-constant spectrum down to the 100 -kHz range and up to the megahertz range. Being uniquely defined within an arbitrary constant, each of the high- and low-frequency KK reconstructions was made to honor the simulated dielectric values at a specific frequency to secure a unique constant in the transformation.

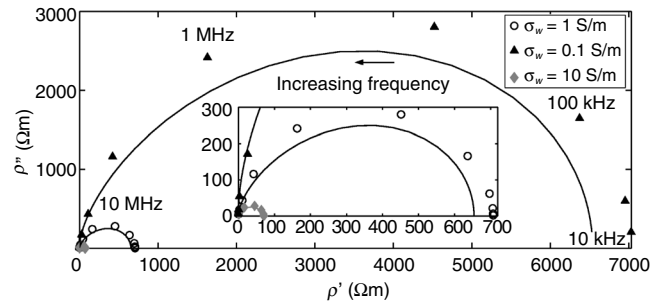


Figure 7. Argand plot of the simulation results described in Figure 6 (markers) together with the corresponding EMA results (plain curves). The insert enlarges the scale of the main plot for values of ρ' smaller than 700 Ωm .

The accuracy of our numerical simulations of effective electrical properties is therefore quantified (1) by the fit between the dielectric-constant spectrum directly simulated on each map with respect to that predicted by the KK relations and (2) by the overlap between high- and low-frequency KK predictions. Figure 8 shows the KK results reconstructed from the effective conductivities simulated for maps A, B, and D honored at 1 GHz for the high-frequency KK range and at 300 kHz for the low-frequency KK range. Results verify Criterion 1 accuracy above for all of the pore maps. Criterion 2 is honored for the pore maps in maps B and D with almost perfect overlap between 1 and 60 MHz, while for map A it remains approximate with a slight shift between high- and low-frequency KK reconstructions. Nonetheless, the slopes of the reconstructed dielectric spectra correctly match those of the simulated ones. Overall, the KK relations support the accuracy and reliability of our numerical results.

Finite volume scaling

The (2D) volume considered in our pore-scale numerical model constitutes a portion of a macroscopic continuum. Unlike EMTs that are based on unbounded, homogeneous macroscopic media, our disc-shape numerical rock/water mixtures do not extend beyond a circular boundary. This finite geometry raises two questions with respect to the physical validity of the pore map to represent the behavior of binary mixtures properly: (1) Do the circular boundaries of the pore maps affect the simulation results? (2) Is there a minimum size for the pore map to be representative of the medium?

The first issue was a primary concern in constructing the model, as the circular shape minimizes boundary fringing for homogeneous discs. Inspection confirmed that the internal fields at the boundary of

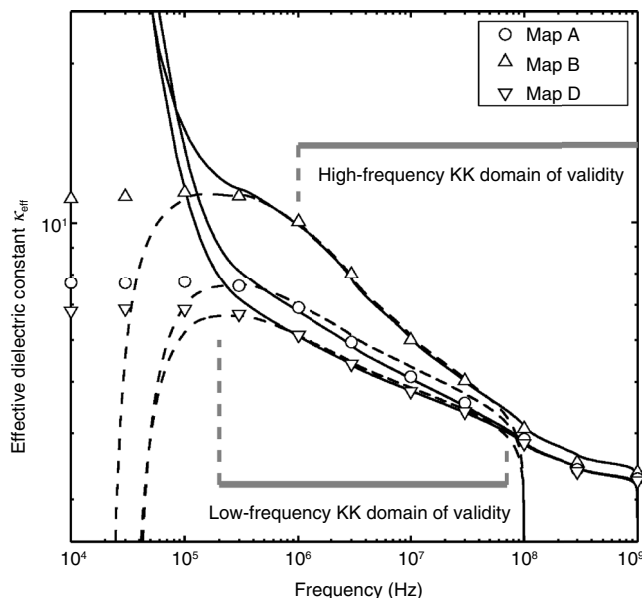


Figure 8. Comparison between the dielectric dispersions simulated for maps A, B, and D in Figure 6, and the Kramers-Kronig (KK) low- and high-frequency predictions reconstructed from the simulated conductivity dispersions. Specific additive constants for the KK reconstructions are defined at 300 kHz in the low-frequency regime (dashed curves) and at 1 GHz in the high-frequency regime (plain curves).

the heterogeneous maps used in this paper are in the worst case affected only marginally by the vicinity of the circular rock/water disc boundary. The most noticeable case of boundary effect was encountered in homogeneous discs; even then, as discussed in Appendix A, only the few outer pixel layers of the disc were affected by the proximity of the boundary.

The issue of scaling is approached with a binary map E (Figure 9), which includes the same 4×48 -pixel brine ellipses and the same 1- μm resolution as discussed previously, but whose size doubles in each direction to 801×801 pixels. Figure 5 compares the simulation results for map E to those of maps A–D and to their best-fit EMA and power-law curves. Dispersions of κ_{eff} and σ_{eff} simulated for map E fit in the range determined by the 401×401 -pixel maps. This comparison indicates that a map size of 401×401 pixels is sufficient to capture the macroscopic geometry of such an 8% porosity medium with inclusions smaller than one-eighth of the map diameter.

Impact of clustering and connectivity

If relative pore geometrical arrangement justifies the differences between the dispersions shown in Figure 4, it is remarkable that no substantial difference exists between the EM dispersions resulting from pores that tend to be regrouped (maps A and C) and those from pores that are more homogeneously distributed in space (maps B and D). To quantify a possible impact of clustering and associated pore connectivity, we consider pore maps with the same pore shapes, sizes, and locations, and we enforce variable degrees of explicit connectivity between these pores.

We start from map D and progressively decrease, then increase, pore connectivity to construct the maps shown in Figure 10. In map D0, we first removed the four connections existing between the ellipses of map D whose edges overlap and leave four more crossovers between other ellipses. In map D00, we also remove these crossovers so that no overlap exists between the ellipses. In maps D1–D3, we increasingly connect the brine inclusions from map D. While map D includes only four pairs of crossed ellipses out of 18 ellipses, map D1 includes three clusters of four connected ellipses each plus two ellipse crossovers. In map D2, a cluster of 11 interconnected

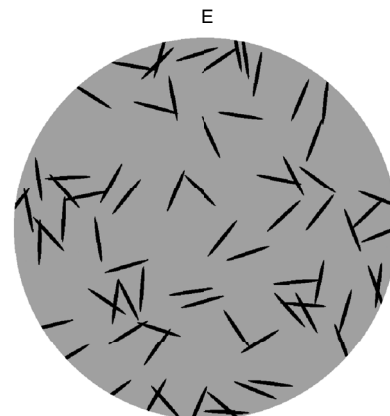


Figure 9. Pore map exhibiting 8% porosity constructed with 801×801 pixels. The brine inclusions consist of 4×48 pixels water ellipses (black) embedded in rock host (gray) consistent with all other pore maps described in this paper.

ellipses is formed along with another cluster of four ellipses. Finally, in map D3, all ellipses are connected except two at the right-hand-side medium boundary. All of these modifications have marginal impact on both the total porosity and the aspect ratio of individual ellipses.

Figure 11 shows the corresponding simulated effective EM dispersions along with EMA results previously obtained for $l = 0.82$ and 0.776 . Note that the spectra simulated for maps D, D0, and D00 exhibit extremely similar behavior. As soon as clustering increases in map D1, the amplitude of dielectric constant at low frequency increases from 6.3 to 9 but the conductivity remains almost identical to that of map D at 1.5 mS/m. Even close to percolation (map D2), low-frequency conductivity doubles compared to that of maps D–D00 to 2 mS/m. On the other hand, the asymptotic dielectric constants on the intervals of 10–100 kHz and 100 MHz to 1 GHz remain practically identical to those of map D1; the only difference between these dispersions appears in the megahertz range where map D2 exhibits more sigmoid curvature than map D1. Maps D1 and D2 are the only maps for which the low-frequency calculated dielectric constant is equal to the EMA prediction for $l = 0.776$.

This result is quite surprising because the cases involved are those where the brine ellipses overlap into nonellipsoidal clusters. Only when percolation is reached in map D3 does the conductivity increase substantially over the entire frequency range and does the dispersion of dielectric constant reach the maximum amplitude with $\epsilon_{\text{eff}} = 15$ below 300 kHz. We verify in Figure 12 that the KK reconstructions of the dielectric spectra of maps D0, D1, and D3 are compatible with our simulation results. The excellent agreement between the two methods gives physical credence to our numerical simulation results and to the corresponding conclusions.

DISCUSSION

Porosity and scaling

More information can be gathered from higher-porosity simulations. In similar fashion to maps A–D, we generated maps F–I as 401×401 -pixel random distributions of 20% porosity pore maps including 4×8 -pixel brine inclusions. To test the relevance of scaling at 20% porosity, we also constructed a 801×801 -pixel map with similar ellipses, map J. Figure 13 shows the corresponding pore maps F–J, and Figure 14 describes the simulation results along with EMA fits. As in the 8% porosity cases, simulation results shown in

Figure 14 emphasize the difference among EM spectra obtained for maps F–J as well as the fact that sigmoids capture average trends for a given porosity and host permittivity.

However, Figure 14 also suggests two technical issues that did not arise with the lower-porosity cases. First, EMA results for $l = 0.82$ still agree well with the simulation results for σ_{eff} in the frequency spectrum but not for κ_{eff} below 10 MHz. Between 10 kHz and 10 MHz, the value $l = 0.875$ better fits the simulated values of κ_{eff} . However, the same value of l does not fit the conductivity data. Thus, unlike in the 8% porosity case, at 20% porosity, it is impossible to reproduce the wide-band electrical dispersion of these 2D rock models with a single value of l and the EMA theory. This observation agrees with the fact that such mixing laws assume finite dilutions of impurities in hosts (i.e., brine inclusions in rock) and confirms the inaccuracy of extrapolations of mixing laws to explain the behavior of nondilute mixtures. Despite this inconsistency of the EMA at 20% porosity, it is remarkable that the mixing-law volumetric model of equation 16 agrees well with the simulation results at 20% porosity; the agreement is as good as the one previously obtained at 8% porosity.

A second item of discussion addresses scaling. At 8% porosity, results for 801×801 -pixel maps agreed with those obtained for 401×401 -pixel maps and also agreed with EMA predictions on average. At 20% porosity, however, the results for the 801×801 -pixel map J hardly fall in the range determined by the results obtained with the 401×401 -pixel realizations, and the simulations could not fit EMA conductivity dispersions. Nevertheless, 801×801 -pixel simulations provide a conductivity dispersion that is closer to EMA predictions than the results obtained with the 401×401 -pixel maps. This suggests that as porosity increases, the representative volume of the micromodel must increase to incorporate large-scale conductivity. Such a behavior would need to be confirmed with 3D simulations where the full extent of interpore connections would be more accurately represented. Despite this scale inaccuracy, it is remarkable that KK relations do confirm the consistency and reliability of the simulation results shown in Figure 15.

Del Río and Whitaker (2000a, 2000b) consider the coupled electric/magnetic fields averaging on both a local-length scale (where most variations take place) and a macroscopic-length scale dictated by the long-distance interactions between the composite elements. The parallel with our simulation results is remarkable and allows us to probe the range of those length scales. At 8% porosity, the consistency of the results obtained for pore maps measuring 401×401 and 801×801 pixels suggests that a length scale of 401 pixels (eight times the elliptical pore size) is sufficient to capture both local and microscopic effects. At 20% porosity, however, a length scale of 801 pixels (16 times the pore size) is not sufficient. This question re-

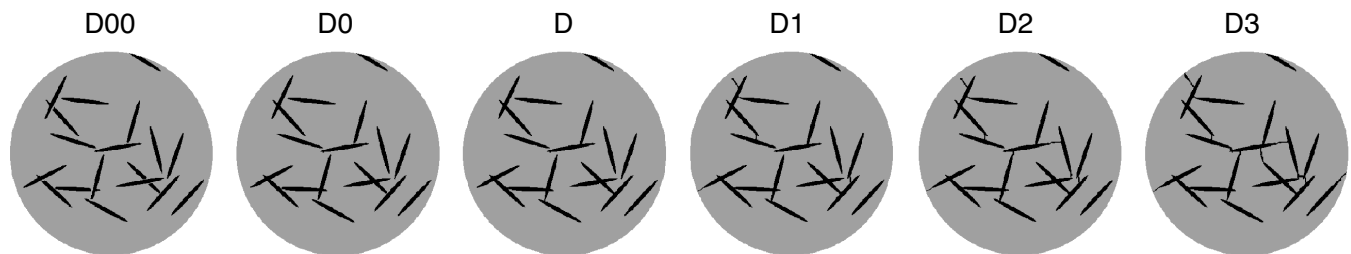


Figure 10. Pore maps modified after map D and exhibiting increasing connectivity between brine pores. Brine inclusions (black pixels) are embedded in the rock host (gray pixels).

quires more study specifically directed toward the interaction between inclusion density, inclusion size, and appropriate length scale that define a homogeneous composite.

Pore connectivity

The effect of pore connectivity on dielectric dispersion confirmed at 8% porosity is quite striking. Unlike what Lysne's (1983) model suggests, the inclusion shape distribution will not be the only parameters in the EM wide-band response of the medium. Instead, the degree of connectivity of these shapes, more than their arrangement, is the primary effect on dielectric dispersion. The difference obtained between the dielectric spectra simulated for maps D1 and D2 is also remarkable and suggests that near percolation is a condition to reach single-inflection sigmoid shapes for the frequency dispersion of a dielectric constant. Comparison to other curves shown in Figure 11 supports this argument, as all the dielectric spectra of nonpercolating maps exhibit several inflection points between 1 and 10 MHz.

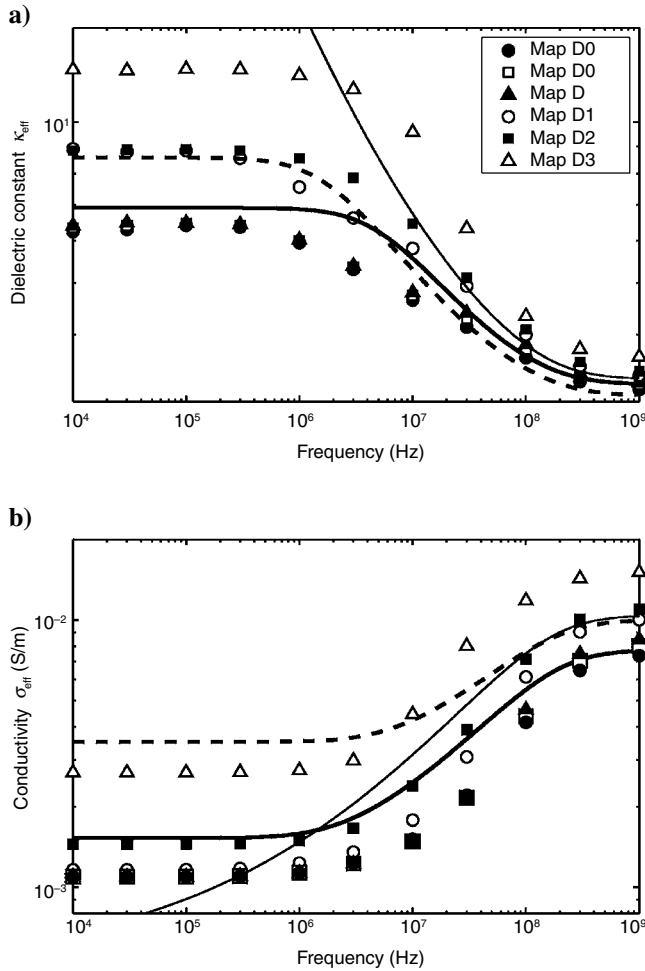


Figure 11. Frequency dispersions of (a) effective dielectric constant and (b) electrical conductivity for pore maps D00 to D3. Simulation results are identified with markers for $\kappa_{\text{host}} = 2$. Thick curves identify the EMA results from equation 14 for $l = 0.82$ (plain curves) and $l = 0.776$ (dashed curves); thin curves identify the power-law results obtained from equation 16.

By contrast, dielectric dispersions simulated for the 20% porosity maps F–J (Figure 14) essentially exhibit one inflection point, which could be associated with the fact that these maps nearly percolate. However, given the scaling issue at higher porosities, more modeling effort is needed to establish conclusive relationships between unicity of the dielectric spectrum inflection and connectivity of the pores. The objective of such 3D simulations should be to quantify a possible relationship between the dielectric spectrum shape and large-scale pore connectivity, i.e., macroscopic hydraulic permeability in a rock sample.

Generalization

An approach similar to the homogeneous cylinder method could be used to estimate effective anisotropic tensorial electrical properties of the pore map. To that end, one could use semianalytical scattering closed-form models for homogeneous anisotropic cylinders (Monzon and Damaskos, 1986; Wu and Yasumoto, 1997) and invert the values of electric fields and currents as the effective conductivity tensor and effective permittivity tensor for the pore map in equation 6.

The 2D assumption for the rock models we considered allowed us to simulate frequency-dispersion curves with relatively modest computer resources. Specifically, the FFT-MoM method allowed us to simulate frequency dispersion of 2D pore maps as large as one megapixel in a desktop workstation, whereas 3D computer simulations require larger resources currently unavailable to us. We chose to study TM polarization to probe the electric and dielectric contrasts within the 2D medium; however, the 3D response of disordered composites will be formed by unknown combinations of both TM

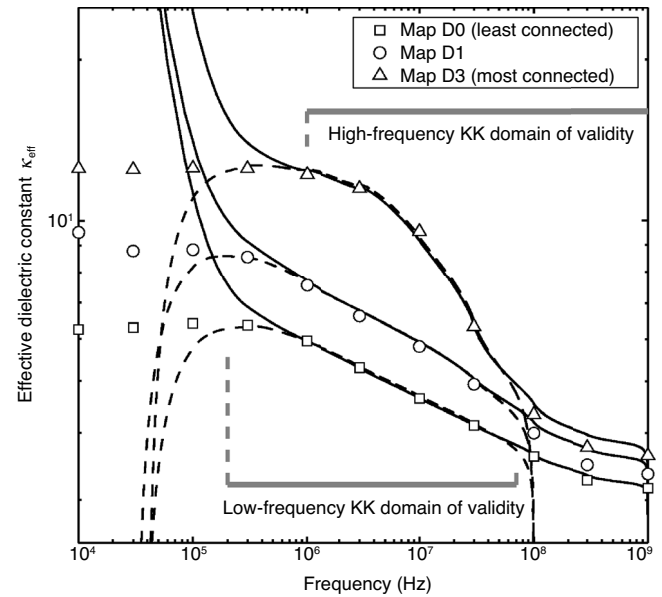


Figure 12. Comparison between the dielectric dispersions simulated for maps D0, D1, and D3 in Figure 11 and their KK low- and high-frequency predictions reconstructed from the simulated conductivity dispersions. Additive constants for the KK reconstructions are defined at 300 kHz in the low-frequency regime (dashed curves) and at 1 GHz in the high-frequency regime (plain curves).

and transverse electric (TE) polarization responses. Three-dimensional simulations can be performed based on the concepts presented in this paper using the closed-form field solution for homogeneous isotropic or anisotropic cylinders and the minimization of the

cost function Ψ_3 in equation 13 on the entire simulation volume instead of the cross section of the 2D model. These 3D simulations will provide accurate estimation of low-frequency effective properties and will quantify cross-polarization effects in 3D pore geometries.

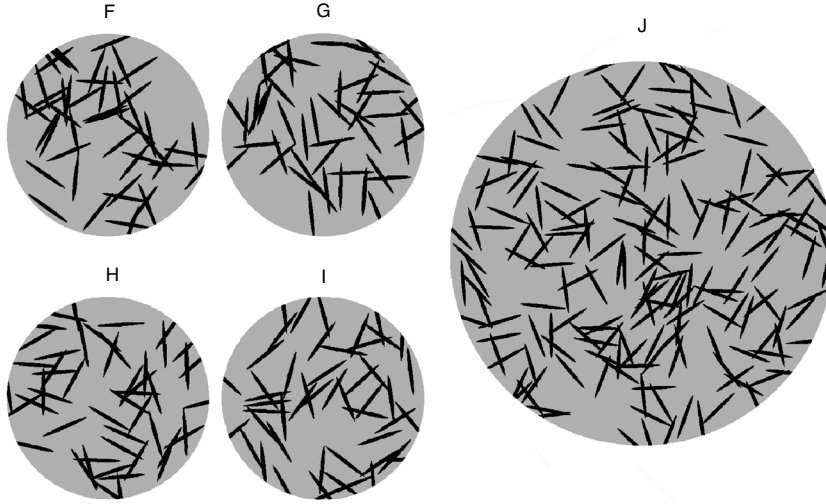


Figure 13. Pore maps exhibiting 20% porosity constructed with 401×401 pixels (F–I) and 801×801 pixels (J). Brine inclusions consist of 4×48 pixels water ellipses (black) embedded in rock host (gray), consistent with all other pore maps described in this paper.

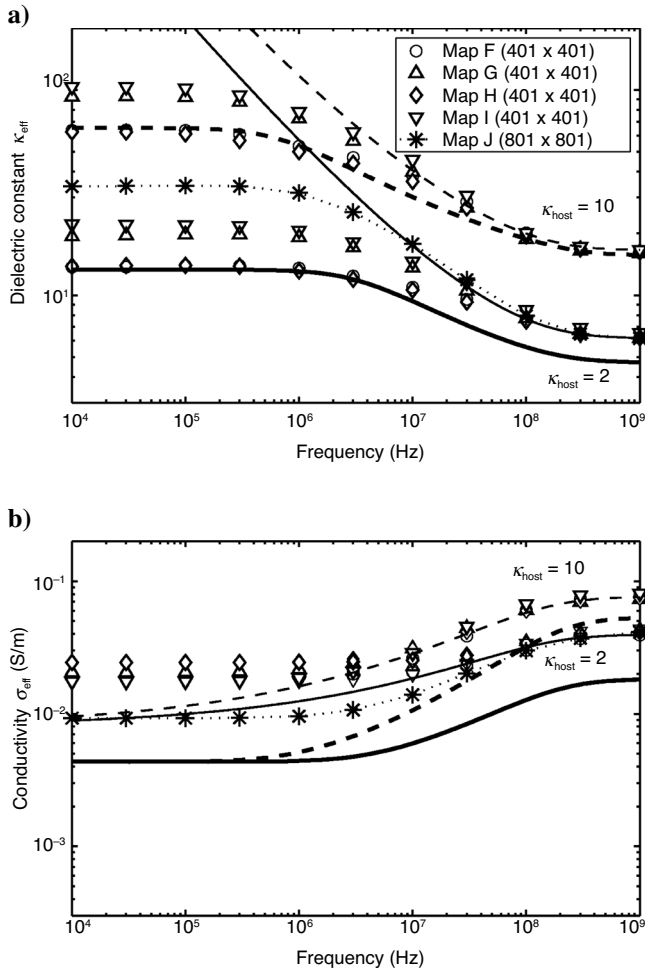


Figure 14. Frequency dispersion of (a) effective dielectric constant and (b) electrical conductivity for pore maps F–J. Simulation results are identified with markers for $\kappa_{\text{host}} = 2$ and $\kappa_{\text{host}} = 10$ (for map J, $\kappa_{\text{host}} = 2$ only). Corresponding mixing-law results are represented with plain curves for $\kappa_{\text{host}} = 2$ and dashed curves for $\kappa_{\text{host}} = 10$. Thick curves identify the EMA results from equation 14 that best fit both dielectric dispersions simultaneously (for $l = 0.875$); thin curves identify the power-law results from equation 16.

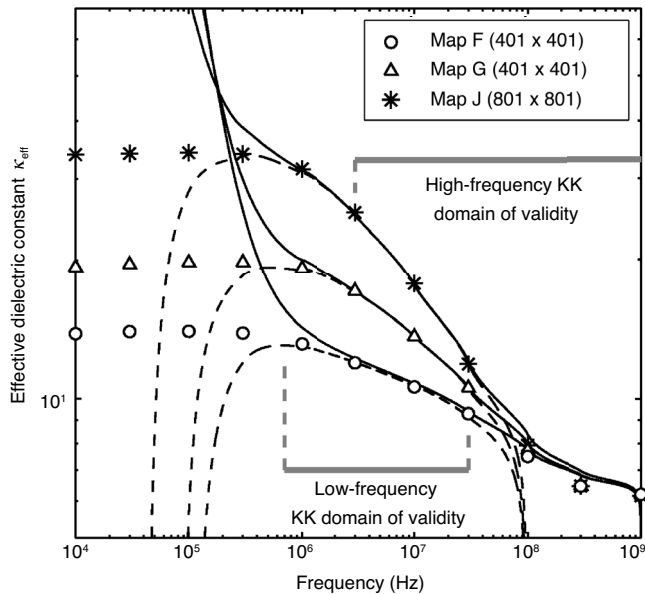


Figure 15. Comparison between the dielectric dispersions simulated for maps F, G, and J in Figure 14 and the KK low- and high-frequency predictions reconstructed from the simulated conductivity dispersions. Anchors for the KK reconstructions are defined at 1 MHz in the low-frequency regime (dashed curves) and at 1 GHz in the high-frequency regime (plain curves).

CONCLUSIONS

We introduce a new method to quantify the influence of pore and grain geometry on the kilohertz-gigahertz EM response of saturated rocks. In so doing, we calculate wide-band dispersion curves of effective electrical conductivity and dielectric constant for explicit arbitrary rock model geometries. Unlike existing numerical methods based on reflection/transmission coefficient calculations at the medium boundary, our method remains valid on the entire kilohertz-gigahertz frequency range and is verified to honor the KK relations.

Effective medium theories only describe the average wide-band polarization of rocks for given porosities and pore eccentricities. In general, the wide-band electrical behavior of each rock departs from an average sigmoid curve by a factor of two for relatively low values of pore eccentricity (12 in our simulations), depending on pore clustering and connectivity. Although usually neglected by existing dielectric mixing laws, we show that clustering and connectivity have a significant impact on wide-band EM dispersions.

Scaling is a major issue for the pore-scale modeling of homogeneous properties such as conductivity and dielectric constant at all frequencies. Our work suggests a relationship between single inflection of the dielectric spectrum and brine percolation across the model rock; however, larger-scale simulations in two and three dimensions are needed to establish quantitative relationships among dielectric spectrum modes, large-scale pore connectivity, and the presence of isolated pore structures in the rock skeleton.

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APPENDIX A

ACCURACY OF THE NUMERICAL SOLUTION OF THE ELECTRIC FIELD INTEGRAL EQUATION WITH A 2D FFT-MOM METHOD

We consider a homogeneous disc-shaped scatterer and analyze the numerical error between internal electric fields calculated with 2D FFT-MoM and internal fields calculated analytically with a cylindrical expansion. The numerical approach discretizes the disc on a grid of 401×401 pixels, i.e., the disc has a radius equal to 200 pixels. We vary the frequency logarithmically from 10 kHz to 1 GHz. Electrical parameters used for this comparison are as follows: $\sigma_w = 1$ S/m, $\kappa_w = 70$, $\sigma_{\text{disc}} = 0.01$ S/m, and $\kappa_{\text{disc}} = 20$. The diameter of the disc is only 401 μm , which makes the analytical value for the internal electric field constant over the entire disc surface.

In what follows, the projections of the numerical error for each pixel i are defined as

$$\text{error}_{x,z}^i = \frac{E_{x,z}^i - E_{x,z}^a}{E_{x,z}^a}, \quad (\text{A-1})$$

where $E_{x,z}^i$ are the Cartesian projections of the internal electric field calculated for pixel i , and $E_{x,z}^a$ are the projections of the analytical internal electric field calculated close to the disc center. Figure A-1 describes the results obtained from this analysis. For all tested frequencies, Figure A-1a shows that the numerical error within the disc lies within 10^{-2} or 1% of the analytical solution. The peripheries of the white circles (zones of maximal error) illustrate the increasing importance of fringing effects for decreasing values of frequency. We use two norms, the infinity norm L_{∞} (maximal relative error amplitude) and the L_1 norm (mean relative error amplitude), to measure the overall numerical error achieved at each frequency.

Figure A-1b shows the results obtained by computing the errors over the complete or part of the scatterer body. A disc radius of 200 pixels comprises the entire disc discretized on the 401×401 -pixel grid, whereas smaller concentric discs ignore the outmost pixels of the scattering disc where fringing occurs. The L_1 norm yields less than 1% error at all frequencies regardless of fringing effects. On the other hand, the L_{∞} norm achieves a similar accuracy when computed over the central 180-pixel disc, but it shows the degradation of field homogeneity at the disc edge when it is computed for larger radii.

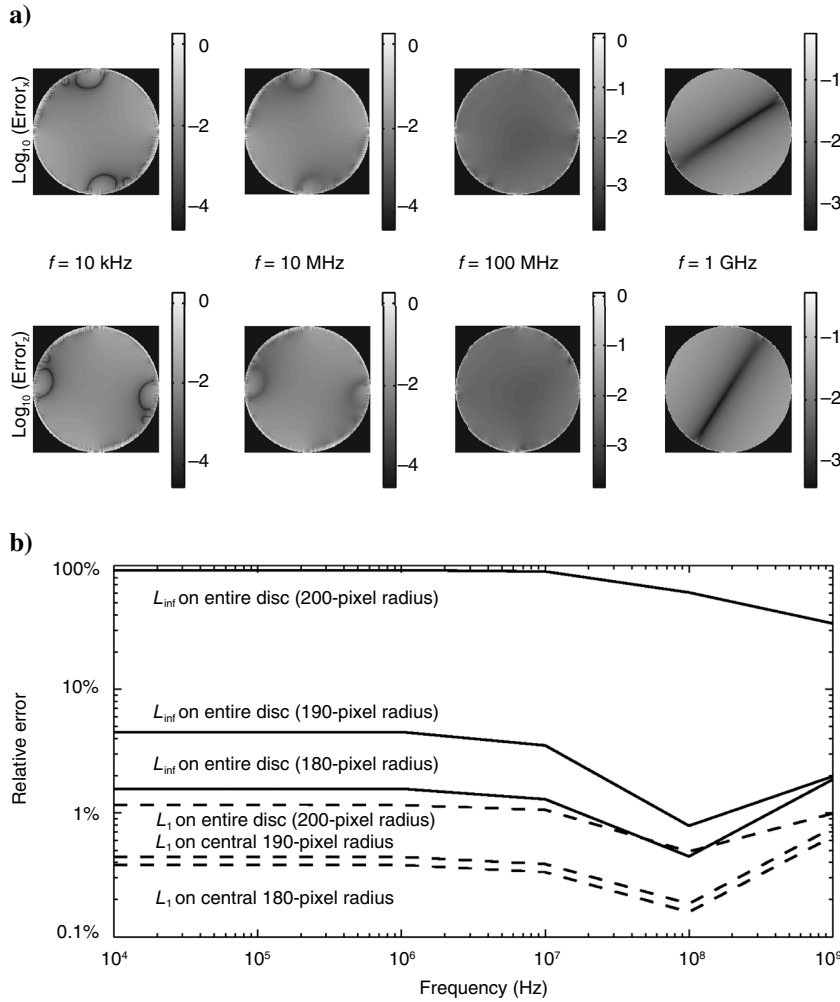


Figure A-1. Analysis of 2D FFT-MoM numerical error for the internal electric fields within a homogeneous cylinder of 401×401 -pixel resolution. (a) Distribution of the error for both x and z projections at several frequencies. (b) L_{∞} and L_1 norms of the relative error computed as a function of frequency for decreasing values of disc radius within the scatterer.

APPENDIX B

PRACTICAL IMPLEMENTATION OF THE MINIMIZATION PROBLEM

Numerical experiments indicate that the estimated values of effective electric conductivity and dielectric constant obtained from minimizing the cost function defined by equation 12 are not always consistent with the expected asymptotic behavior at low and high frequencies. To circumvent this problem, we first define the electrical field and electric-current residuals as

$$\Delta \mathbf{E} = \mathbf{E} - \mathbf{E}_h \quad (\text{B-1})$$

and

$$\Delta \mathbf{J} = \bar{\sigma}^* \mathbf{E} - \bar{\sigma}_h^* \mathbf{E}_h, \quad (\text{B-2})$$

respectively. Moreover, we note that, according to Schwartz's inequality,

$$\begin{aligned} \Psi &= \|\Delta \mathbf{J} - \bar{\sigma}_b^* \Delta \mathbf{E}\|_{L_2} \\ &\leq \|\Delta \mathbf{J}\|_{L_2} + |\bar{\sigma}_b^*| \times \|\Delta \mathbf{E}\|_{L_2}. \end{aligned} \quad (\text{B-3})$$

We now compare the performance of three choices of cost function when inverting the conductivity and dielectric constant of a homogeneous disc-shaped pore map. These cost functions are defined as

$$\Psi_1 = \|\Delta \mathbf{J} - \bar{\sigma}_b^* \Delta \mathbf{E}\|_{L_2}, \quad (\text{B-4})$$

$$\begin{aligned} \Psi_2 &= \|\text{Re}(\Delta \mathbf{J} - \bar{\sigma}_b^* \Delta \mathbf{E})\|_{L_2} \\ &\quad + \|\text{Im}(\Delta \mathbf{J} - \bar{\sigma}_b^* \Delta \mathbf{E})\|_{L_2}, \end{aligned} \quad (\text{B-5})$$

and

$$\Psi_3 = \|\Delta \mathbf{J}\|_{L_2} + |\bar{\sigma}_b^*| \times \|\Delta \mathbf{E}\|_{L_2}. \quad (\text{B-6})$$

To assess the performance of the three cost functions, we consider combinations of three values of conductivity σ_{disc} (0.001, 0.01, and 0.1 S/m) and three values of dielectric constant κ_{disc} (20, 200, and 2000) assigned to a homogeneous disc-shaped pore map. We compute the internal electric fields for frequencies ranging from 1 kHz to 1 GHz and use the three cost functions described by equations B-4 to B-6 to invert the effective properties σ_{eff} and κ_{eff} , starting from the same forward simulation results. In all cases, the background brine consists of 1 S/m conductivity and a dielectric constant equal to 70. Results from this exercise are shown in Figure B-1.

When σ_{disc} is at least equal to 0.1 S/m, both σ_{eff} and κ_{eff} are estimated properly with the three cost functions, with slightly better results for the case of cost function Ψ_3 compared with Ψ_1 and Ψ_2 . Incorrect results are obtained when κ_{disc} is as large as 2000 above 3 MHz; fortunately, this case is not realistic in saturated rocks. When σ_{disc} is lower than 0.1 S/m, Ψ_1 overestimates κ_{eff} at low frequencies and systematically yields wrong (even negative) values for σ_{eff} . Cost function Ψ_2 yields relatively better results and always leads to the right values of κ_{eff} , but the inverted values σ_{eff} are as inconsistent as those obtained with Ψ_1 . Cost function Ψ_3 yields the most accurate and stable results in all the cases. As indicated in Figure B-1, bounds exist beyond which κ_{eff} and/or σ_{eff} are not properly inverted with cost function Ψ_3 .

For realistic cases of rock/fluid mixtures, one can state conservatively that the dielectric constant is never higher than 50 above 1 MHz. In this case, we expect the minimization algorithm to converge for conductivities above 0.01 S/m and for frequencies as high as 1 GHz. Similarly, we expect the minimization algorithm to converge for conductivities above 0.001 S/m and for frequencies as high as 10 MHz.

In the presence of low-frequency enhancement resulting from clays or other electrochemical factors, the dielectric constant might take on a value of several thousands below 1 MHz. In this case, both

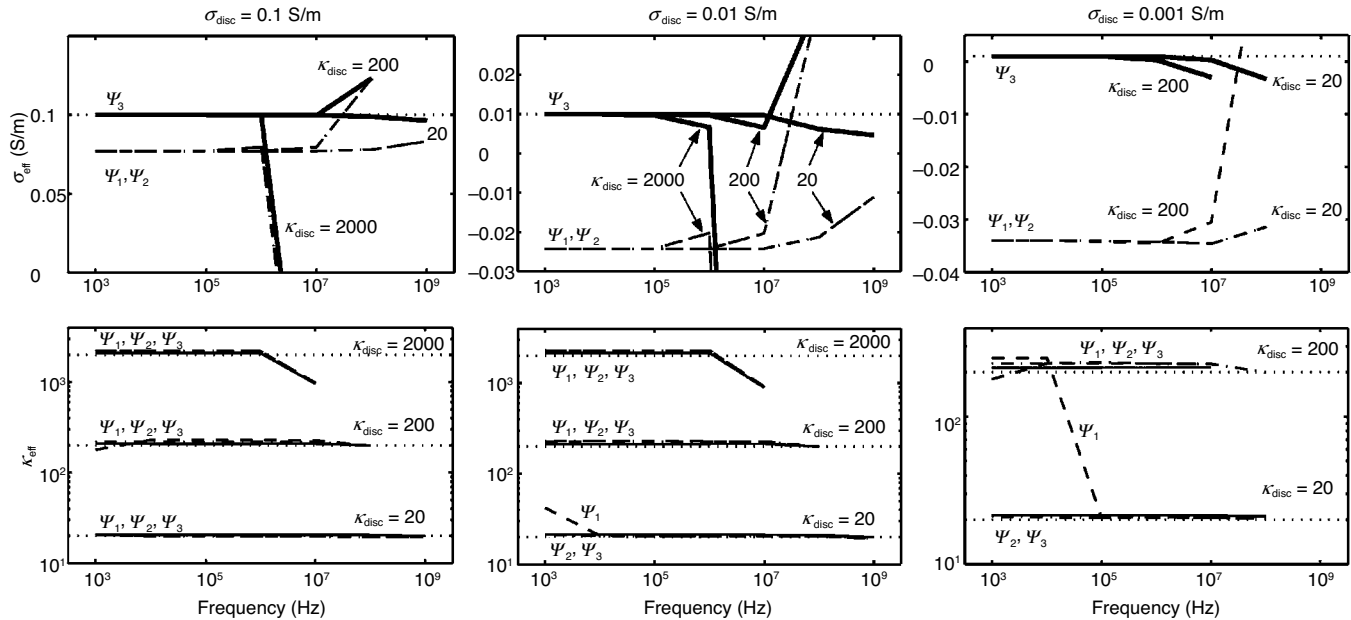


Figure B-1. Inversion results for a 401×401 -pixel homogeneous disc for different combinations of effective conductivity σ_{disc} and dielectric constant κ_{disc} with different cost functions. Dash-dotted lines denote use of cost function Ψ_1 , dashed lines denote Ψ_2 , and plain lines denote Ψ_3 . Dotted lines indicate the true values for σ_{disc} and κ_{disc} .

dielectric constant and conductivity will be correctly inverted for conductivity values as low as 0.01 S/m.

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